

Builder Basics — The Sloped Roof (Extension · Beginner)

Topic: Variables · **Course:** Builder Basics · **Type:** Extension Activity · **Level:** Beginner · **Time:** about 30 minutes

 This is a bonus challenge, not a main lesson. Try it once you are happy building with loops.

1 · Predict

Open the **Sloped Roof** world. Do not run it yet.

It stacks 4 square **rings** of stairs, each on top of the last. Each ring is **shorter** than the one below, so the roof rises to a peak.

What shape does this make? Circle one: flat square · stepped pyramid roof · single line

Why is each ring smaller than the one below? Circle one: to step up to a peak · to get wider · to dig down

2 · Watch the Variable Change

A **variable** is a name that holds a number. It can **change** while the program runs.

Here it is `roof_len`. It starts at 7 and loses 2 each ring.

So the four rings are 7, 5, and then... Circle one: 3, 1 · 4, 2 · 6, 5

The widest ring is at the... Circle one: bottom · top

3 · Think It Through

If `roof_len` stayed 7, what would the roof look like? Circle one: a flat box, all rings the same size
· a pointed peak · a deep hole

Which number makes the roof start wider at the bottom? Circle one: the starting 7 · the peak
block · the world name

4 · Show Your Work 📷 🎥

1. Open the **Sloped Roof** world.
2. Type `slope` in the Minecraft chat and press enter to run the program.
3. Watch the rings build up, getting smaller each time, until they reach the peak.

Then change the starting number. Make `roof_len` start at **9** instead of 7, run it again, and see how the roof changes.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Say these out loud, filling in the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was ____.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.